

EX NO : 3

DATE : 22.06.26

DATA ENGINEERING

AIM:

Data Preprocessing and Encoding for Employee Dataset

PROBLEM STATEMENT:

You have been provided with an employee dataset (emp-data.xlsx) containing both numerical and categorical attributes related to employee demographics, roles, performance, and compensation.

Perform the following preprocessing tasks:

1. Identify Categorical and Numerical Attributes
2. Apply Label Encoding to binary/ordinal categorical columns (Gender, Education)
3. Apply One-Hot Encoding to nominal categorical columns (Department, City, Work_Mode, Job_Role)
4. Apply StandardScaler to normalize numerical attributes
5. Apply MinMaxScaler to scale numerical attributes to [0,1] range
6. Visualize the effect of scaling using histograms and boxplots
7. Compare Label Encoding and One-Hot Encoding on the Department column

DATASET:

emp-data.xlsx — Employee data with attributes: Employee_ID, Name, Age, Gender, Education, Department, City, Work_Mode, Job_Role, Experience, Salary, Performance_Score

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder, StandardScaler, MinMaxScaler

# Load Dataset
df = pd.read_excel("emp-data.xlsx")
print("===== ORIGINAL DATASET =====")
print(df.head())

# Task 1: Identify Categorical and Numerical Attributes
print("\n===== TASK 1 =====")
numerical_columns = df.select_dtypes(include=['int64', 'float64']).columns
categorical_columns = df.select_dtypes(include=['object']).columns
print("\nNumerical Attributes:")
print(list(numerical_columns))
print("\nCategorical Attributes:")
print(list(categorical_columns))

# Task 2: Label Encoding
print("\n===== TASK 2 =====")
label_encoder = LabelEncoder()
df_label = df.copy()
df_label["Gender"] = label_encoder.fit_transform(df_label["Gender"])
df_label["Education"] = label_encoder.fit_transform(df_label["Education"])
print(df_label[["Gender", "Education"]].head())

# Task 3: One-Hot Encoding
print("\n===== TASK 3 =====")
```

```

df_onehot = pd.get_dummies(
    df_label,
    columns=["Department", "City", "Work_Mode", "Job_Role"]
)
print(df_onehot.head())

# Task 4: StandardScaler
print("\n===== TASK 4 =====")
standard_df = df_onehot.copy()
standard_scaler = StandardScaler()
columns_to_scale = [
    "Age",
    "Experience",
    "Salary",
    "Performance_Score"
]
standard_df[columns_to_scale] = standard_scaler.fit_transform(
    standard_df[columns_to_scale]
)
print(standard_df[columns_to_scale].head())

# Task 5: MinMaxScaler
print("\n===== TASK 5 =====")
minmax_df = df_onehot.copy()
minmax_scaler = MinMaxScaler()
minmax_df[columns_to_scale] = minmax_scaler.fit_transform(
    minmax_df[columns_to_scale]
)
print(minmax_df[columns_to_scale].head())

# Task 6: Visualization
print("\n===== TASK 6 =====")
plt.figure(figsize=(10,4))
plt.hist(df["Salary"], bins=10)
plt.title("Salary Before Standard Scaling")
plt.xlabel("Salary")
plt.ylabel("Frequency")
plt.show()

plt.figure(figsize=(10,4))
plt.hist(standard_df["Salary"], bins=10)
plt.title("Salary After Standard Scaling")
plt.xlabel("Scaled Salary")
plt.ylabel("Frequency")
plt.show()

plt.figure(figsize=(6,4))
plt.boxplot(df["Salary"])
plt.title("Salary Before Standard Scaling")
plt.show()

plt.figure(figsize=(6,4))
plt.boxplot(standard_df["Salary"])
plt.title("Salary After Standard Scaling")
plt.show()

```

```

print("\nInterpretation:")
print("StandardScaler changes the scale of Salary.")
print("The shape of the distribution remains the same.")
print("The mean becomes approximately 0 and the standard deviation becomes 1.")

# Task 7: Compare Label Encoding and One-Hot Encoding
print("\n===== TASK 7 =====")
department_label = df.copy()
department_label["Department"] = LabelEncoder().fit_transform(
    department_label["Department"]
)
print("\nLabel Encoding:")
print(department_label[["Department"]].head())

department_onehot = pd.get_dummies(
    df[["Department"]],
    columns=["Department"]
)
print("\nOne-Hot Encoding:")
print(department_onehot.head())

```

OUTPUT:

Task 1 – Numerical and Categorical Attributes:

```

===== ORIGINAL DATASET =====
  Employee_ID  Gender  Age  Department  Education  ...  Work_Mode  Job_Role  Experience  Salary  Performance_Score
0      E001    Male   36  Operations    PhD      ...    Remote   Developer      16    85286              8
1      E002    Male   40  Marketing    PG      ...    Onsite    Analyst      7    57956              5
2      E003    Male   31  Marketing    PhD      ...    Onsite  Consultant      7    53872              4
3      E004  Female   42         IT      PG      ...    Hybrid   Developer     17   127276              4
4      E005  Female   29  Operations    UG      ...    Onsite    Analyst      6    64724              5

[5 rows x 12 columns]

===== TASK 1 =====

Numerical Attributes:
['Age', 'Experience', 'Salary', 'Performance_Score']

Categorical Attributes:
['Employee_ID', 'Gender', 'Department', 'Education', 'Marital_Status', 'City', 'Work_Mode', 'Job_Role']

```

Task 2 – Label Encoding (Gender, Education):

```

===== TASK 2 =====
   Gender  Education
0        1         1
1        1         0
2        1         1
3        0         0
4        0         2

```

Task 3 – One-Hot Encoding:

```
===== TASK 3 =====
Employee_ID  Gender  Age  Education  ...  Job_Role_Consultant  Job_Role_Developer  Job_Role_Engineer  Job_Role_Management
0  E001      1   36      1  ...             False             True             False             False
1  E002      1   40      0  ...             False             False            False            False
2  E003      1   31      1  ...             True              False            False            False
3  E004      0   42      0  ...             False             True             False            False
4  E005      0   29      2  ...             False             False            False            False

[5 rows x 26 columns]
```

Task 4 – StandardScaler:

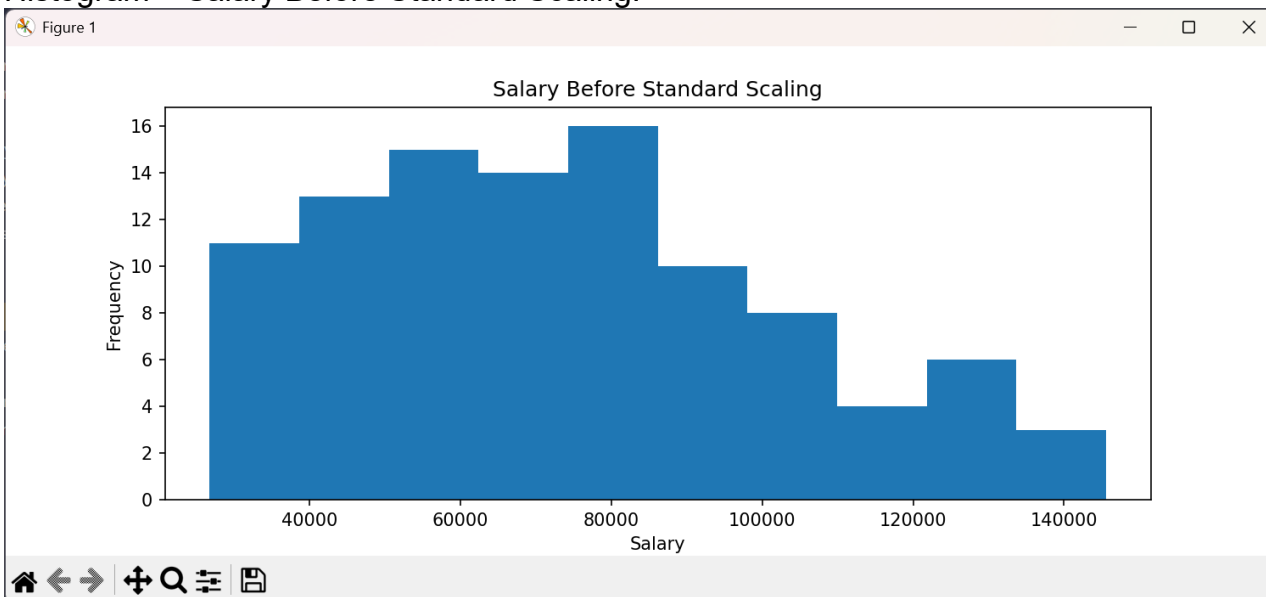
```
===== TASK 4 =====
Age  Experience  Salary  Performance_Score
0 -0.287193    1.019497  0.381884    0.561610
1  0.257249   -0.538308 -0.541412   -0.916312
2 -0.967746   -0.538308 -0.679383   -1.408952
3  0.529470    1.192586  1.800443   -1.408952
4 -1.239968   -0.711397 -0.312767   -0.916312
```

Task 5 – MinMaxScaler:

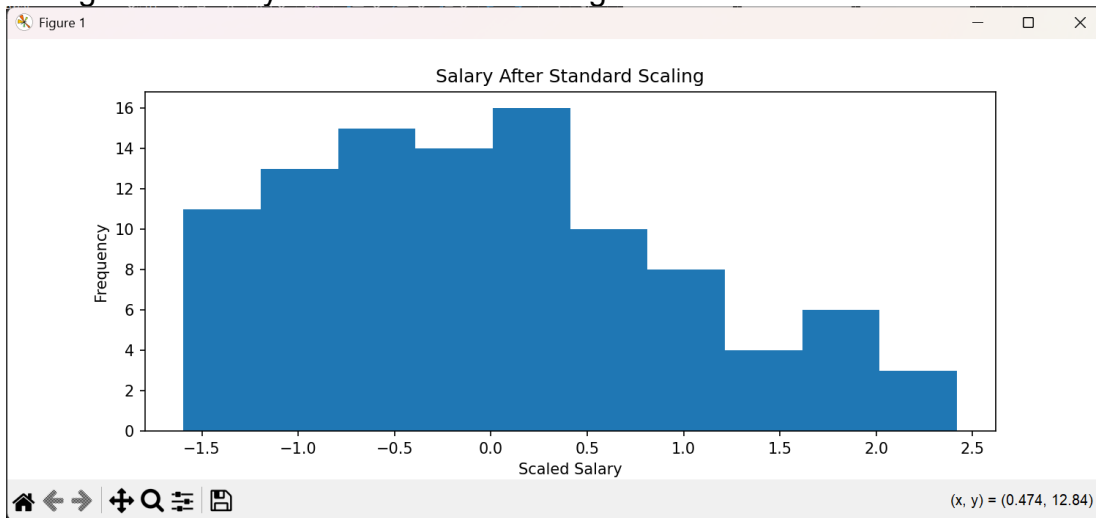
```
===== TASK 5 =====
Age  Experience  Salary  Performance_Score
0  0.40625    0.789474  0.492541    0.666667
1  0.53125    0.315789  0.262601    0.166667
2  0.25000    0.315789  0.228241    0.000000
3  0.59375    0.842105  0.845823    0.000000
4  0.18750    0.263158  0.319544    0.166667
```

Task 6 – Visualization:

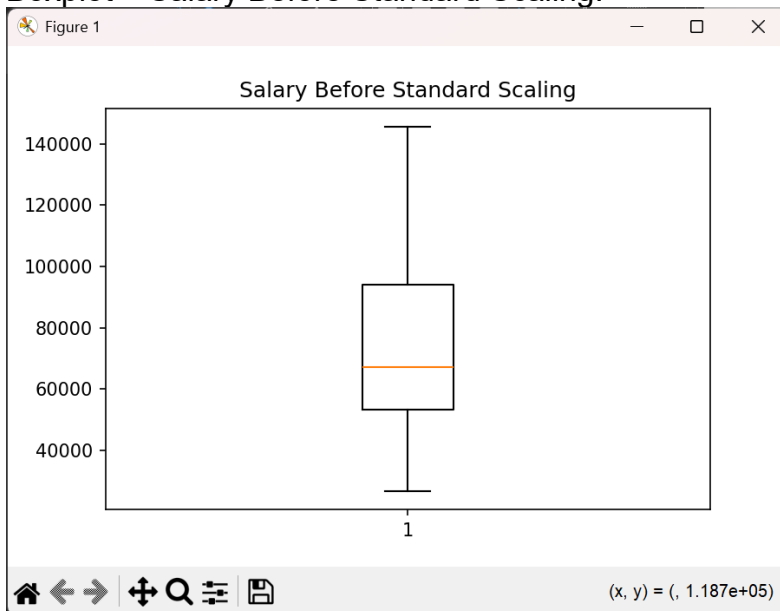
Histogram – Salary Before Standard Scaling:



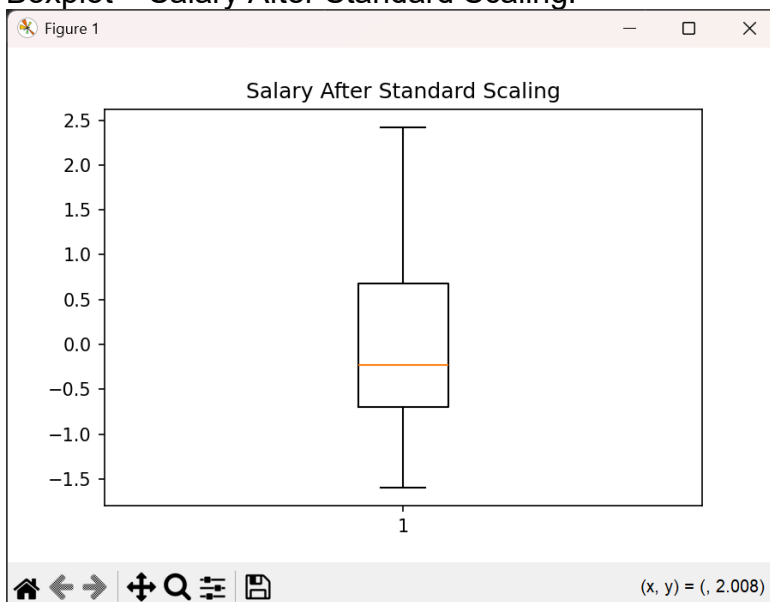
Histogram – Salary After Standard Scaling:



Boxplot – Salary Before Standard Scaling:



Boxplot – Salary After Standard Scaling:



Interpretation:

StandardScaler changes the scale of Salary. The shape of the distribution remains the same. The mean becomes approximately 0 and the standard deviation becomes 1.

Task 7 – Label Encoding vs One-Hot Encoding (Department):

Label Encoding:

Department	
0	4
1	3
2	3
3	2
4	4

One-Hot Encoding:

	Department_Finance	Department_HR	Department_IT	Department_Marketing	Department_Operations
0	False	False	False	False	True
1	False	False	False	True	False
2	False	False	False	True	False
3	False	False	True	False	False
4	False	False	False	False	True

RESULT:

The employee dataset was successfully preprocessed. Categorical attributes were encoded using Label Encoding and One-Hot Encoding. Numerical attributes were scaled using StandardScaler and MinMaxScaler. The effect of scaling was visualized using histograms and boxplots, confirming that StandardScaler normalizes the distribution while preserving its shape.